

[71]

Seat No: _____



No. of printed pages: 2

SARDAR PATEL UNIVERSITY
Master of Information Technology M Sc IT Semester II Examinations
PS02CINT53 – Artificial Intelligence

Time: 10:00 a.m. to 01:00 p.m.

Date: 12/11/2022, Saturday

Marks: 70

Q1. Choose the most appropriate option for each question. [8]

- 1 Forward chaining is a strategy used by _____.
(A) Inference engine (B) Control system (C) Turing test (D) None of these
- 2 _____ is a type of knowledge.
(A) Tacit (B) Implicit (C) Meta knowledge (D) All of these
- 3 _____ set does not have boundaries.
(A) Logical (B) Rough (C) Fuzzy (D) Nibble
- 4 A fuzzy membership degree always results in _____.
(A) Boolean values (B) -1 to 1 (C) Any value (D) 0 to 1
- 5 Which of the following is a model of artificial neural network?
(A) Perceptron (B) Forward pass (C) Backward pass (D) None of these
- 6 A nucleus of a neural network consists of _____ function.
(A) Soma (B) Activation (C) Fitness (D) None of these
- 7 Quality of GA solution mainly depends on _____.
(A) Encoding (B) Operation (C) Fitness function (D) None of these
- 8 GA is used when _____.
(A) Solution space is large (B) Traditional solutions not available
(C) Both A and B (D) None of these

Q2. Answer the following questions (Any Seven): [14]

- 1 Give two examples of KBS in education domain.
- 2 Define the Turing test.
- 3 Define crisp set and fuzzy set.
- 4 Define production system.
- 5 Draw fuzzy membership function of high temperature and low temperature of a typical air conditioner machine.
- 6 Define fixed increment perceptron learning.
- 7 List any two activation functions in ANN.
- 8 List any four encoding strategies in GA.
- 9 List stopping criteria of GA.

①

C.P.T.O.)

Q3. Answer the following questions:

- A Write a short note on knowledge acquisition process in KBS. [6]
B Explain water jug problem using parameters of your choice. [6]

OR

- B Explain frame knowledge representation in detail by giving suitable example. [6]

Q4. Answer the following questions:

- A Explain fuzzy control system. [6]
B Write a short note on fuzzy relation. Also provide suitable examples and operations on fuzzy relations. [6]

OR

- B List and explain different operations on fuzzy sets in detail with examples. [6]

Q5. Answer the following questions:

- A Explain in detail a Hopfield model of ANN. [6]
B Discuss how a perceptron solves linearly separable problems by giving an example. [6]

OR

- B Design a neural network to select a mobile based on the following example training data. [6]

Sr. No.	X1 Brand	X2 Price	X3 Memory	X4 OS	X5 Camera	O1 M1	O2 M2	O3 M3
1	0.9	0.9	0.6	0.7	0.8	0	1	0
2	0.8	0.6	0.8	0.7	0.9	0	0	1
3	0.5	0.5	0.7	0.6	0.7	1	0	0

Q6. Answer the following questions:

- A List and discuss genetic operators namely (i) selection, (ii) mutation, and (iii) crossover in detail by giving example. [6]
B Write a short note on how GA can be used to solve a single variable function optimization problem. [6]

OR

- B Write a short note on notion and applications of schema in GA. [6]

X

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